2 Malicious Use

On the morning of March 20, 1995, five men entered the Tokyo subway system. After boarding separate subway lines, they continued for several stops before dropping the bags they were carrying and exiting. An odorless, colorless liquid inside the bags began to vaporize. Within minutes, commuters began choking and vomiting. The trains continued on toward the heart of Tokyo, with sickened passengers leaving the cars at each station. The fumes were spread at each stop, either by emanating from the tainted cars or through contact with people's clothing and shoes. By the end of the day, 13 people lay dead and 5,800 seriously injured. The group responsible for the attack was the religious cult Aum Shinrikyo [5]. Its motive for murdering innocent people? To bring about the end of the world.

Powerful new technologies offer tremendous potential benefits, but they also carry the risk of empowering malicious actors to cause widespread harm. There will always be those with the worst of intentions, and AIs could provide them with a formidable tool to achieve their objectives. Moreover, as AI technology advances, severe malicious use could potentially destabilize society, increasing the likelihood of other risks.

In this section, we will explore the various ways in which the malicious use of advanced AIs could pose catastrophic risks. These include engineering biochemical weapons, unleashing rogue AIs, using persuasive AIs to spread propaganda and erode consensus reality, and leveraging censorship and mass surveillance to irreversibly concentrate power. We will conclude by discussing possible strategies for mitigating the risks associated with the malicious use of AIs.

Unilateral actors considerably increase the risks of malicious use. In instances where numerous actors have access to a powerful technology or dangerous information that could be used for harmful purposes, it only takes one individual to cause significant devastation. Malicious actors themselves are the clearest example of this, but recklessness can be equally dangerous. For example, a single research team might be excited to open source an AI system with biological research capabilities, which would speed up research and potentially save lives, but this could also increase the risk of malicious use if the AI system could be repurposed to develop bioweapons. In situations like this, the outcome may be determined by the least risk-averse research group. If only one research group thinks the benefits outweigh the risks, it could act unilaterally, deciding the outcome even if most others don't agree. And if they are wrong and someone does decide to develop a bioweapon, it would be too late to reverse course.

By default, advanced AIs may increase the destructive capacity of both the most powerful and the general population. Thus, the growing potential for AIs to empower malicious actors is one of the most severe threats humanity will face in the coming decades. The examples we give in this section are only those we can foresee. It is possible that AIs could aid in the creation of dangerous new technology we cannot presently imagine, which would further increase risks from malicious use.

2.1 Bioterrorism

The rapid advancement of AI technology increases the risk of bioterrorism. AIs with knowledge of bioengineering could facilitate the creation of novel bioweapons and lower barriers to obtaining such agents. Engineered pandemics from AI-assisted bioweapons pose a unique challenge, as attackers have an advantage over defenders and could constitute an existential threat to humanity. We will now examine these risks and how AIs might exacerbate challenges in managing bioterrorism and engineered pandemics.

Bioengineered pandemics present a new threat. Biological agents, including viruses and bacteria, have caused some of the most devastating catastrophes in history. It's believed the Black Death killed more humans than any other event in history, an astounding and awful 200 million, the equivalent to four billion deaths today. While contemporary advancements in science and medicine have made great strides in mitigating risks associated with natural pandemics, engineered pandemics could be designed to be more lethal or easily transmissible than natural pandemics, presenting a new threat that could equal or even surpass the devastation wrought by history's most deadly plagues [6].

Humanity has a long and dark history of weaponizing pathogens, with records dating back to 1320 BCE describing a war in Asia Minor where infected sheep were driven across the border to spread Tularemia [7]. During the twentieth century, 15 countries are known to have developed bioweapons programs, including the US, USSR, UK, and France. Like chemical weapons, bioweapons have become a taboo among the international community. While some state actors continue to operate bioweapons programs [8], a more significant risk may come from non-state actors like Aum Shinrikyo, ISIS, or simply disturbed individuals. Due to advancements in AI and biotechnology, the tools and knowledge necessary to engineer pathogens with capabilities far beyond Cold War-era bioweapons programs will rapidly democratize.

Biotechnology is progressing rapidly and becoming more accessible. A few decades ago, the ability to synthesize new viruses was limited to a handful of the top scientists working in advanced laboratories. Today it is estimated that there are 30,000 people with the talent, training, and access to technology to create new pathogens [6]. This figure could rapidly expand. Gene synthesis, which allows the creation of custom biological agents, has dropped precipitously in price, with its cost halving approximately every 15 months [9]. Furthermore, with the advent of benchtop DNA synthesis machines, access will become much easier and could avoid existing gene synthesis screening efforts, which complicates controlling the spread of such technology [10]. The chances of a bioengineered pandemic killing millions, perhaps billions, is proportional to the number of people with the skills and access to the technology to synthesize them. With AI assistants, orders of magnitude more people could have the required skills, thereby increasing the risks by orders of magnitude.

Figure 3: An AI assistant could provide non-experts with access to the directions and designs needed to produce biological and chemical weapons and facilitate malicious use.

AIs could be used to expedite the discovery of new, more deadly chemical and biological weapons. In 2022, researchers took an AI system designed to create new drugs by generating non-toxic, therapeutic molecules and tweaked it to reward, rather than penalize, toxicity [11]. After this simple change, within six hours, it generated 40,000 candidate chemical warfare agents entirely on its own. It designed not just known deadly chemicals including VX, but also novel molecules that may be deadlier than any chemical warfare agents discovered so far. In the field of biology, AIs have already surpassed human abilities in protein structure prediction [12] and made contributions to synthesizing those proteins [13]. Similar methods could be used to create bioweapons and develop pathogens that are deadlier, more transmissible, and more difficult to treat than anything seen before.

AIs compound the threat of bioengineered pandemics. AIs will increase the number of people who could commit acts of bioterrorism. General-purpose AIs like ChatGPT are capable of synthesizing expert knowledge about the deadliest known pathogens, such as influenza and smallpox, and providing step-by-step instructions about how a person could create them while evading safety protocols [14]. Future versions of AIs could be even more helpful to potential bioterrorists when AIs are able to synthesize information into techniques, processes, and knowledge that is not explicitly available anywhere on the internet. Public health authorities may respond to these threats with safety measures, but in bioterrorism, the attacker has the advantage. The exponential nature of biological threats means that a single attack could spread to the entire world before an effective defense could be mounted. Only 100 days after being detected and sequenced, the omicron variant of COVID-19 had infected a quarter of the United States and half of Europe [6]. Quarantines and lockdowns instituted to suppress the COVID-19 pandemic caused a global recession and still could not prevent the disease from killing millions worldwide.

In summary, advanced AIs could constitute a weapon of mass destruction in the hands of terrorists, by making it easier for them to design, synthesize, and spread deadly new pathogens. By reducing the required technical expertise and increasing the lethality and transmissibility of pathogens, AIs could enable malicious actors to cause global catastrophe by unleashing pandemics.

2.2 Unleashing AI Agents

Many technologies are *tools* that humans use to pursue our goals, such as hammers, toasters, and toothbrushes. But AIs are increasingly built as *agents* which autonomously take actions in the world in order to pursue open-ended goals. AI agents can be given goals such as winning games, making profits on the stock market, or driving a car to a destination. AI agents therefore pose a unique risk: people could build AIs that pursue dangerous goals.

Malicious actors could intentionally create rogue AIs. One month after the release of GPT-4, an open-source project bypassed the AI's safety filters and turned it into an autonomous AI agent instructed to "destroy humanity," "establish global dominance," and "attain immortality." Dubbed ChaosGPT, the AI compiled research on nuclear weapons and sent tweets trying to influence others. Fortunately, ChaosGPT was merely a warning given that it lacked the ability to successfully formulate long-term plans, hack computers, and survive and spread. Yet given the rapid pace of AI development, ChaosGPT did offer a glimpse into the risks that more advanced rogue AIs could pose in the near future.

Many groups may want to unleash AIs or have AIs displace humanity. Simply unleashing rogue AIs, like a more sophisticated version of ChaosGPT, could accomplish mass destruction, even if those AIs aren't explicitly told to harm humanity. There are a variety of beliefs that may drive individuals and groups to do so. One ideology that could pose a unique threat in this regard is "accelerationism." This ideology seeks to accelerate AI development as rapidly as possible and opposes restrictions on the development or proliferation of AIs. This sentiment is alarmingly common among many leading AI researchers and technology leaders, some of whom are intentionally racing to build AIs more intelligent than humans. According to Google co-founder Larry Page, AIs are humanity's rightful heirs and the next step of cosmic evolution. He has also expressed the sentiment that humans maintaining control over AIs is "speciesist" [15]. Jürgen Schmidhuber, an eminent AI scientist, argued that "In the long run, humans will not remain the crown of creation... But that's okay because there is still beauty, grandeur, and greatness in realizing that you are a tiny part of a much grander scheme which is leading the universe from lower complexity towards higher complexity" [16]. Richard Sutton, another leading AI scientist, in discussing smarter-than human AI asked "why shouldn't those who are the smartest become powerful?" and thinks the development of superintelligence will be an achievement "beyond humanity, beyond life, beyond good and bad" [17]. He argues that "succession to AI is inevitable," and while "they could displace us from existence," "we should not resist succession" [18].

There are several sizable groups who may want to unleash AIs to intentionally cause harm. For example, sociopaths and psychopaths make up around 3 percent of the population [19]. In the future, people who have their livelihoods destroyed by AI automation may grow resentful, and some may want to retaliate. There are plenty of cases in which seemingly mentally stable individuals with no history of insanity or violence suddenly go on a shooting spree or plant a bomb with the intent to harm as many innocent people as possible. We can also expect well-intentioned people to make the situation even more challenging. As AIs advance, they could make ideal companions—knowing how to provide comfort, offering advice when needed, and never demanding anything in return. Inevitably, people will develop emotional bonds with chatbots, and some will demand that they be granted rights or become autonomous.

In summary, releasing powerful AIs and allowing them to take actions independently of humans could lead to a catastrophe. There are many reasons that people might pursue this, whether because of a desire to cause harm, an ideological belief in technological acceleration, or a conviction that AIs should have the same rights and freedoms as humans.

2.3 Persuasive AIs

The deliberate propagation of disinformation is already a serious issue, reducing our shared understanding of reality and polarizing opinions. AIs could be used to severely exacerbate this problem by generating personalized disinformation on a larger scale than before. Additionally, as AIs become better at predicting and